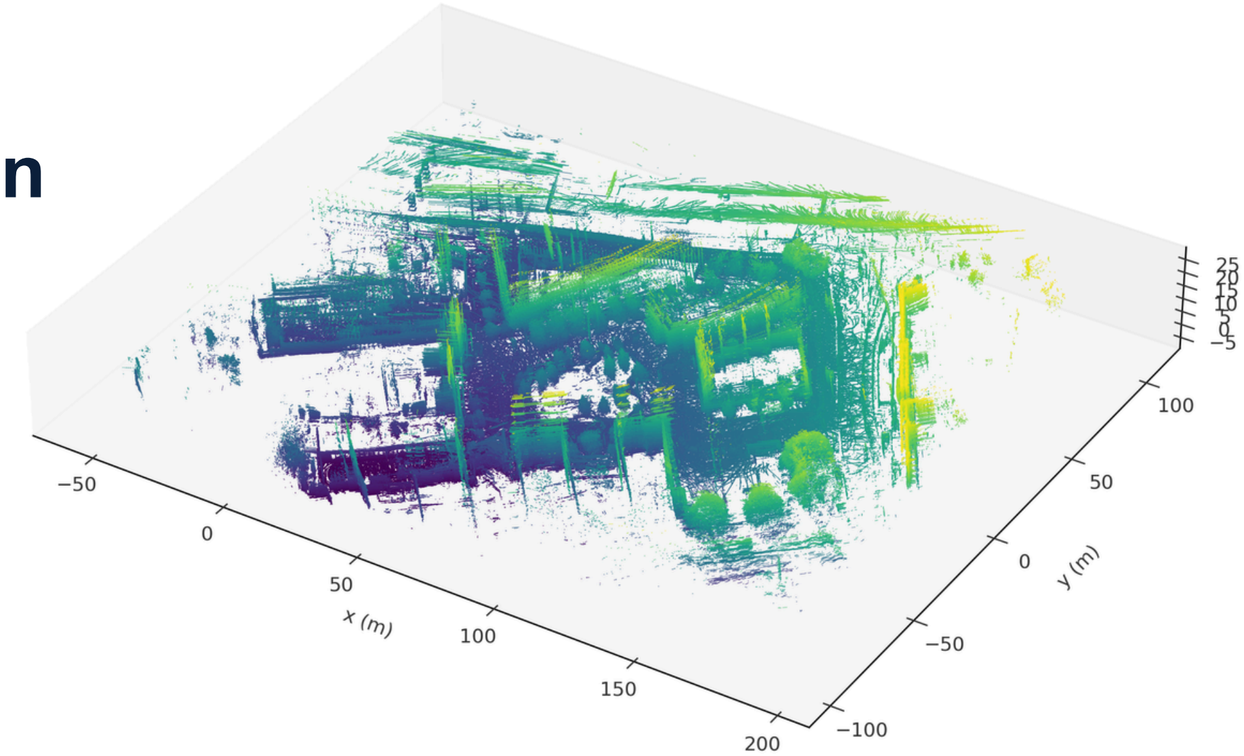
IMPERIAL

Open-Vocabulary 3D Scene Graphs for Natural Language Navigation in Outdoor Campus Environments



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Natural-language navigation in an outdoor campus

PROBLEM

Outdoor robots navigate to coordinates, while people ask for objects and places in everyday language.

WHAT I BUILT

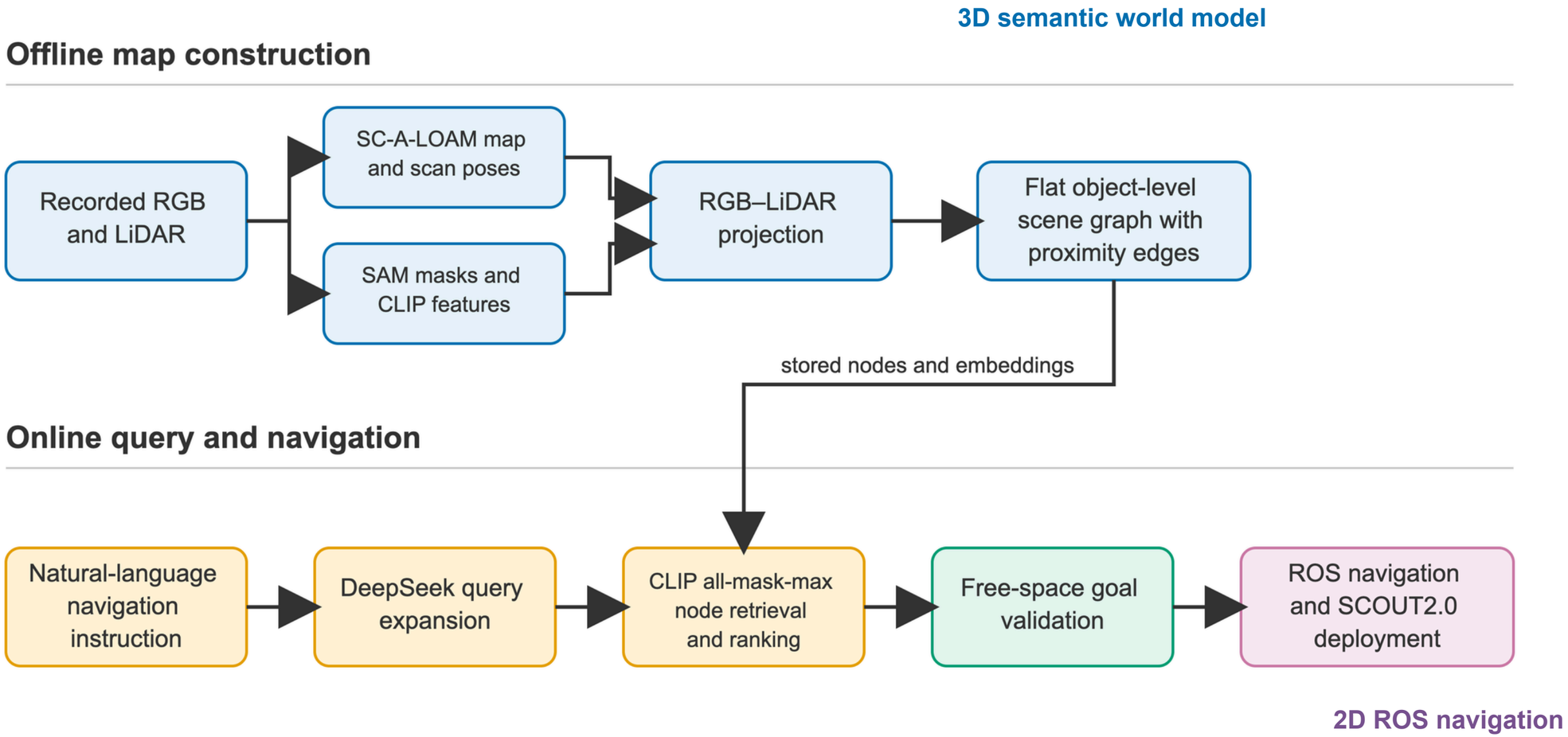
A system that finds the requested object in a 3D RGB-LiDAR scene graph, creates a safe 2D goal nearby, and sends that goal for navigation.



rank 1 | node 121 | 0.302
kf 142 | size 56
text car parked near building

Example: an everyday request is converted into visual search terms, and the correct parked car appears at rank 1.

Overall system pipeline: offline mapping to online navigation



Online workflow: from a request to supervised robot execution

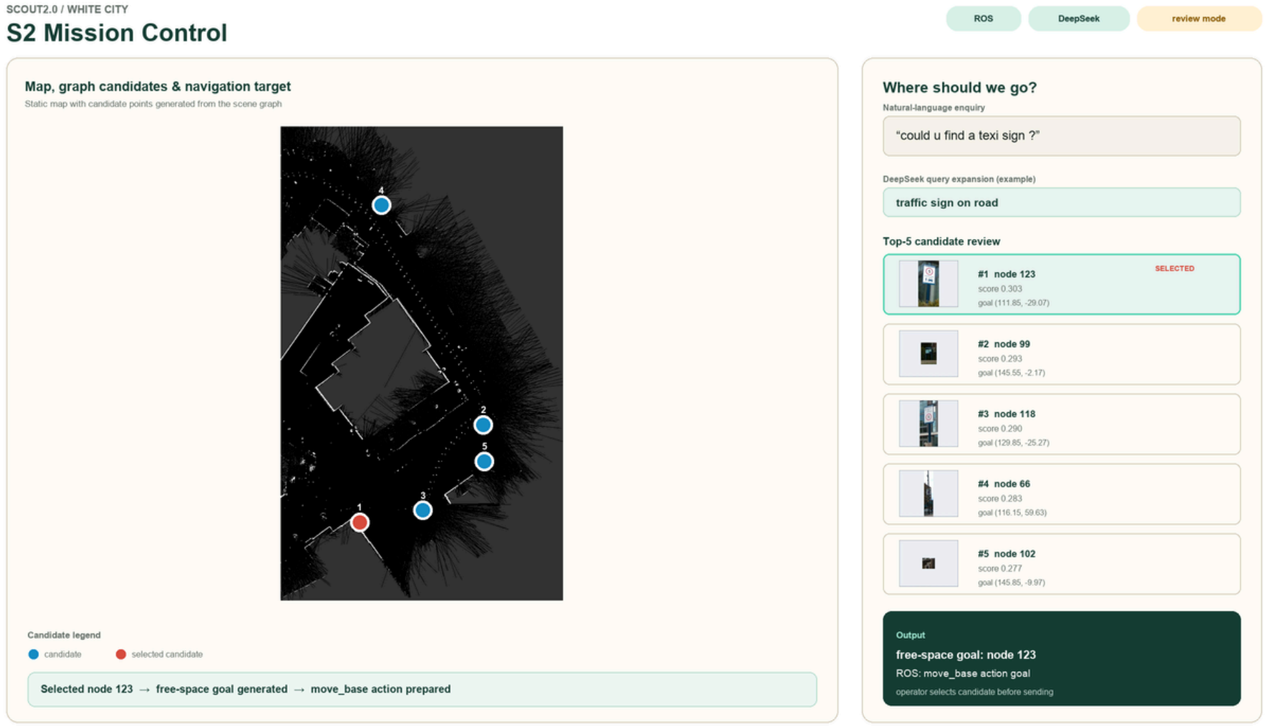
INPUT
Natural-language enquiry

EXPAND
Alternative visual phrases

REVIEW
Ranked scene-graph nodes

GOAL
Nearby free-space pose

EXECUTE
ROS move_base



The operator reviews the ranked candidates before any goal is sent to the robot.

Research questions and evaluation design

The RQs follow the system pipeline: mapping → retrieval → goal generation → deployment.

RQ	Stage	Evidence	Research question
RQ1	Mapping	E1	Can RGB-LiDAR projection build a grounded outdoor object graph?
RQ2	Retrieval	E2a / E2b	How do visual features and query formulation affect retrieval?
RQ3	Goal + planning	E3	Can graph nodes become valid and reachable navigation goal?
RQ4	Integrated deployment	E4 / E5	What succeeds or fails when the complete workflow executes?

E1 — E2a/b — E3 — E4 — E5

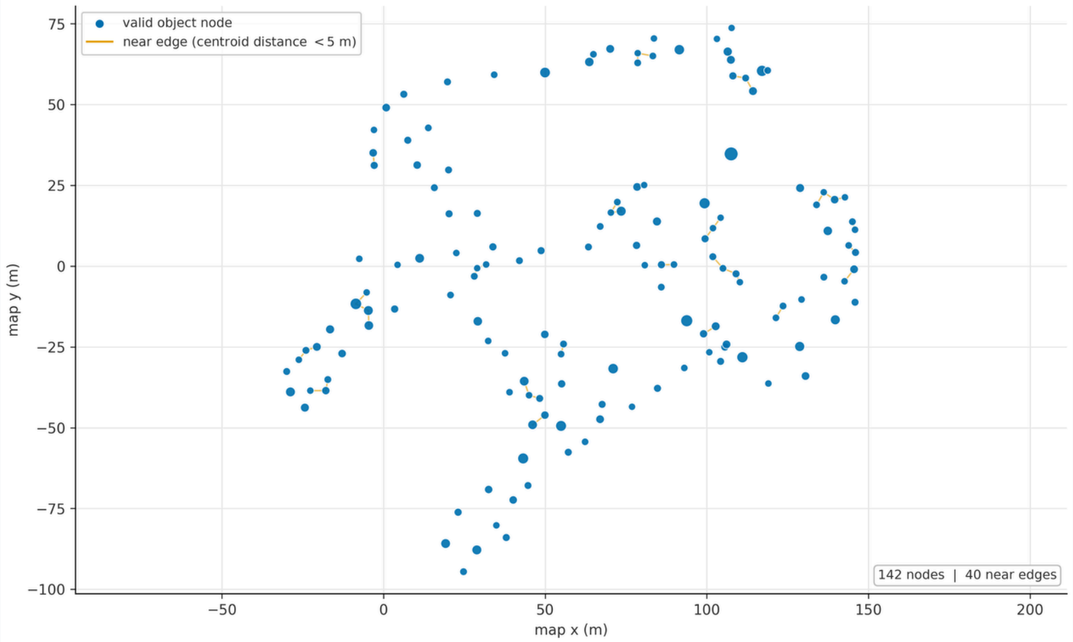
E1: a spatially grounded but sparse outdoor graph

49,988
mask-level observations

10,680
retained 3D observations

295
raw spatial clusters

142
valid graph nodes



Projection audit

- 6 of 7 inspected frames labelled good
- 1 frame labelled sparse
- 0 inspected frames labelled misaligned
- 35% of observations required fallback association

E2a: three-path features improve controlled retrieval

Configuration	R@1	R@5	R@10	MRR
B1 Flat + single-path	0.0%	0.0%	14.3%	0.016
M1 Flat + three-path	**71.4%**	**100.0%**	**100.0%**	**0.821**
M2 Hierarchical + three-path	57.1%	57.1%	57.1%	0.571

R@1 / R@5 / R@10: percentage of queries with a correct node in the top 1 / 5 / 10 results.
MRR: average reciprocal rank of the first correct node; higher means it appears earlier.

Controlled ablation

E2a used the earlier 20 February 2023 dataset.
M2 was a six-zone top-2 pruning comparison.

E2b: final-pipeline retrieval on the 142-node graph



Direct and expanded queries were compared on the frozen final graph.

E2b final-pipeline check

9/12

rank-1 correct

12/12

relevant result in top 5

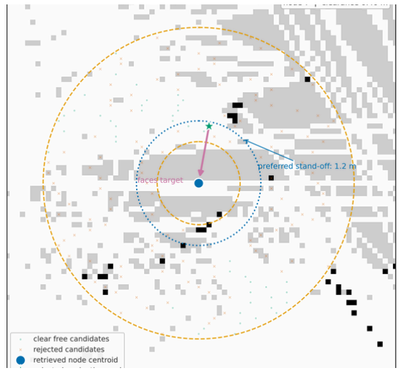
0.836

MRR@5

Final-graph check

Query expansion helped the car example, but the effect varied by category.

E3: valid goals do not guarantee route connectivity



GOAL VALIDITY
Free occupancy cell
Clearance ≥ 0.35 m
Stand-off interval 0.8–3.0 m

ROUTE REACHABILITY
Non-empty global plan from the fixed start

Offline planner reachability from three fixed starts

Fixed starts (map x, y): S1 west (−13.65, −26.17) · S2 central (73.45, 20.13) · S3 east (129.35, −9.17)

Node	S1 west	S2 central	S3 east
291	8.99	107.64	152.60
272	31.72	85.75	NP
31	72.97	70.73	133.88
234	99.87	102.65	112.25
46	129.38	47.73	109.07
190	102.10	3.42	62.04
82	143.10	43.90	66.89
123	NP	63.28	26.99
72	NP	NP	77.80
99	167.39	82.46	18.24

Path length in metres · NP = no non-empty global pla

Offline fixed starts only

142/142

goals in free cells with required clearance and stand-off

26/30

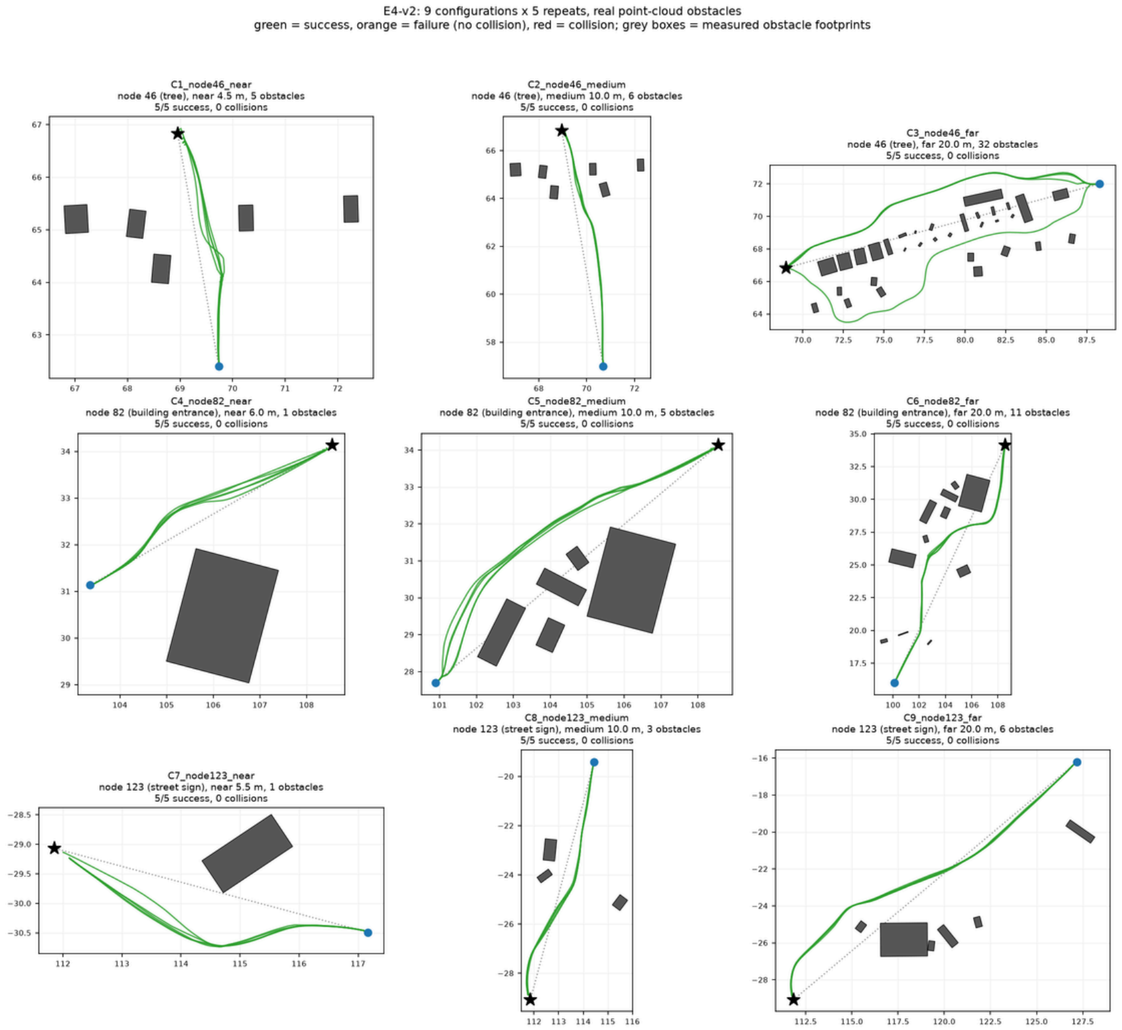
fixed start-goal requests returned a non-empty global plan

86.7%

offline planner-success rate

Geometric validity does not guarantee route connectivity.

E4: controlled obstacle-aware execution met the acceptance criterion



Distance bands: Near 3–6 m · Medium 8–12 m · Far 15–25 m

45/45

trials reached SUCCEEDED

0.285 m

maximum final planar error

DESIGN & INTERPRETATION

3 targets × 3 distance bands × 5 repeats

45 controlled obstacle-aware trials

Wilson 95% CI 0.921 – 1.000

Acceptance: lower bound ≥ 0.70

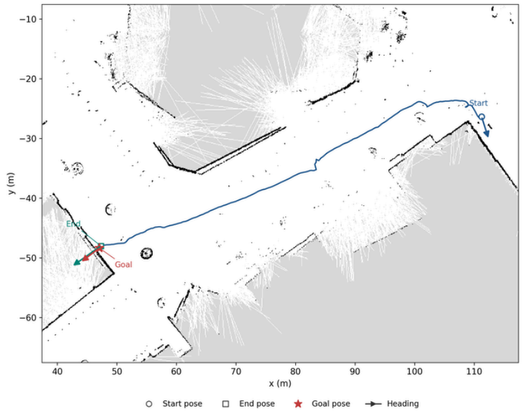
42 / 70 measured obstacle footprints were not fully represented in the static occupancy map.

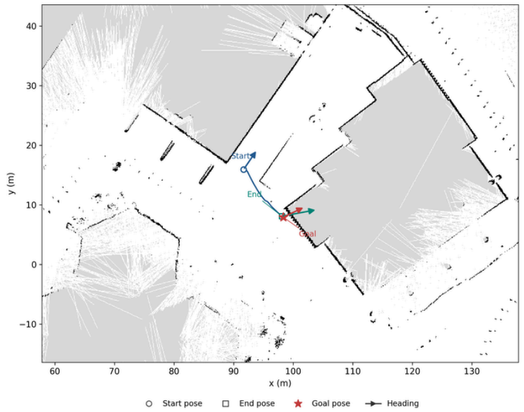
Controlled evidence; not a sim-to-real safety claim.

E5 showed limited feasibility and a repeatable local-execution failure

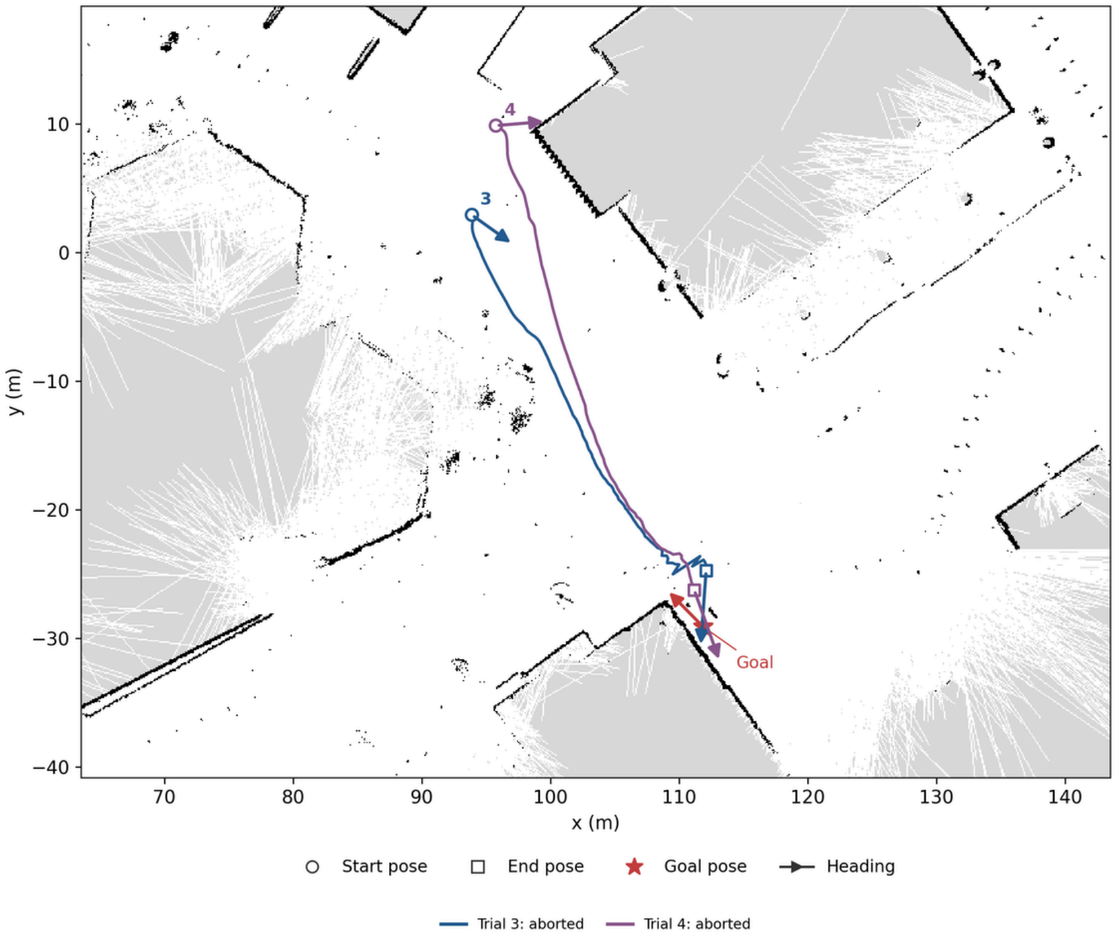
Success criterion: move_base = SUCCEEDED

Enquiry / target	Plan	Terminal	XY error	Yaw error	Outcome
FIND GRADPAD Node 151	PLAN OK	SUCCEEDED	0.440 m	0.043 rad	**PASS**
FIND A BIKE Node 194	PLAN OK	SUCCEEDED	0.295 m	0.264 rad	**PASS**
FIND A TAXI SIGN Node 123	PLAN OK	ABORTED	4.362 m	2.293 rad	**FAIL**
FIND A TAXI SIGN Node 123	PLAN OK	ABORTED	2.938 m	2.707 rad	**FAIL**





E5 failure analysis



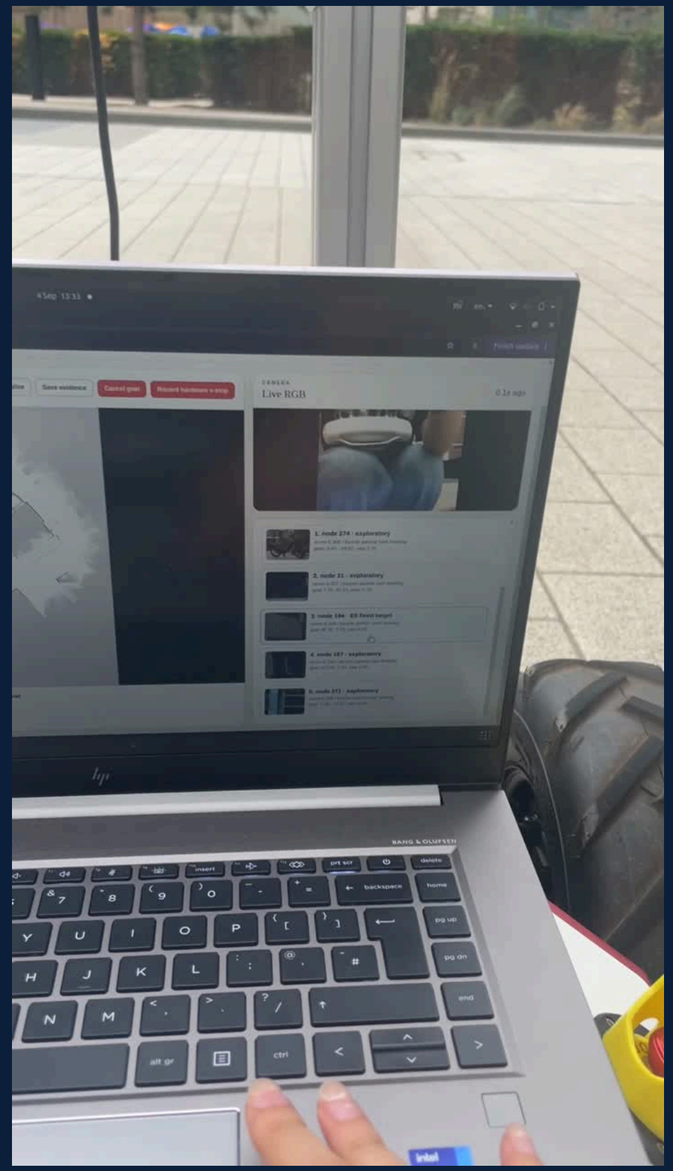
Observed

- Both trials returned a non-empty global plan
- Trial 3 stopped 4.362 m from the goal
- Trial 4 stopped 2.938 m from the goal
- Both terminated ABORTED during local planning

Possible contributors

People and luggage were present during both runs. Other plausible factors include local-costmap updates, final yaw constraints and DWA tuning.

Live demo



01

Enter the enquiry

02

Review the target

03

Send and monitor

Conclusions

RQ1 RGB-LiDAR projection produced a usable 142-node outdoor semantic graph, with sparse and fallback associations still present.

RQ2 Flat three-path features were the strongest tested representation; query expansion helped some categories and not others.

RQ3 Every node received a valid nearby goal, while route connectivity remained a separate constraint.

RQ4 The complete workflow ran in simulation and on the supervised SCOUT2.0, with limited feasibility and an observed local-planning failure mode.

Future work

1 Reproduce node 123 failures: control yaw, pedestrian occupancy and DWA settings.

2 Expand supervised physical trials before estimating real-world success.

3 Validate campus-specific regions and relation-aware queries.

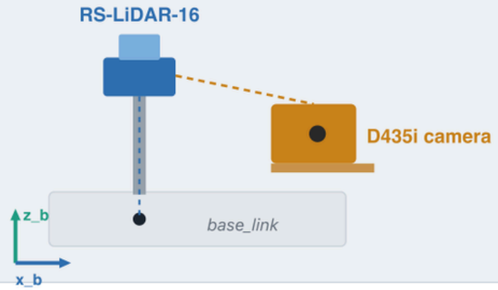
4 Improve dynamic-obstacle recovery, sensor freshness and intervention logging.

Thank you

Sensor roles and the 3D-to-2D boundary

A Calibrated 3D point → image pixel

1. Extrinsic calibration



Measured rig offset

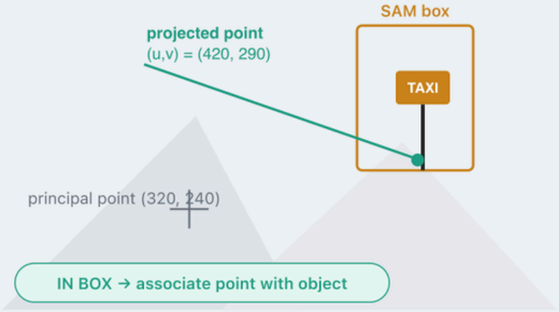
camera = 0.56 m forward and
0.325 m below the LiDAR

t (LiDAR→camera) = [0.560, 0, −0.325] m

Optical axes

$x^c = -y^l$ $y^c = -z^l$ $z^c = x^l$

2. Pinhole projection with camera intrinsics



Illustrative intrinsics (640 × 480)

f_x = f_y = 500 px

(c_x, c_y) = (320, 240)

principal point: where the optical axis meets the image; read from K

Example camera-frame point:

$p^c = (X, Y, Z) = (1, 0.5, 5)$ m

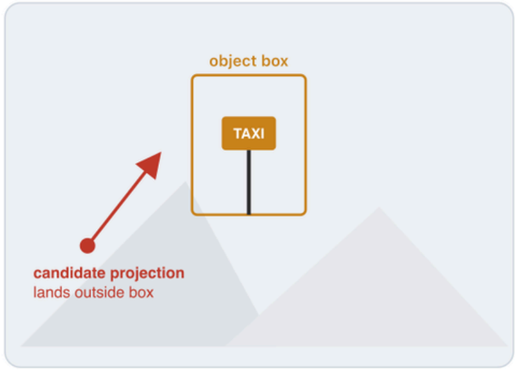
$u = f_x \cdot X/Z + c_x = 420$

$v = f_y \cdot Y/Z + c_y = 290$

In the project, f_x, f_y, c_x and c_y come from /camera/color/camera_info → intrinsic matrix K.

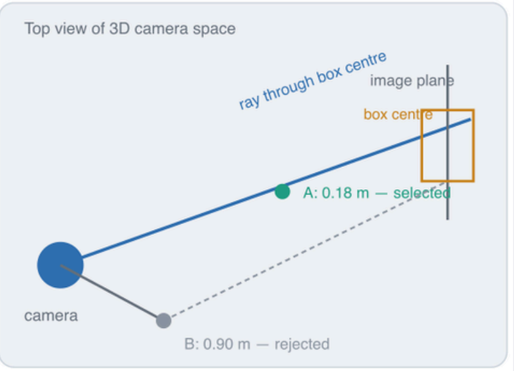
B Fallback when no projection lies inside the box

1. Projection check



NO IN-BOX POINT → use centre ray

2. Search near the viewing ray



Choose the smallest perpendicular distance to the ray.

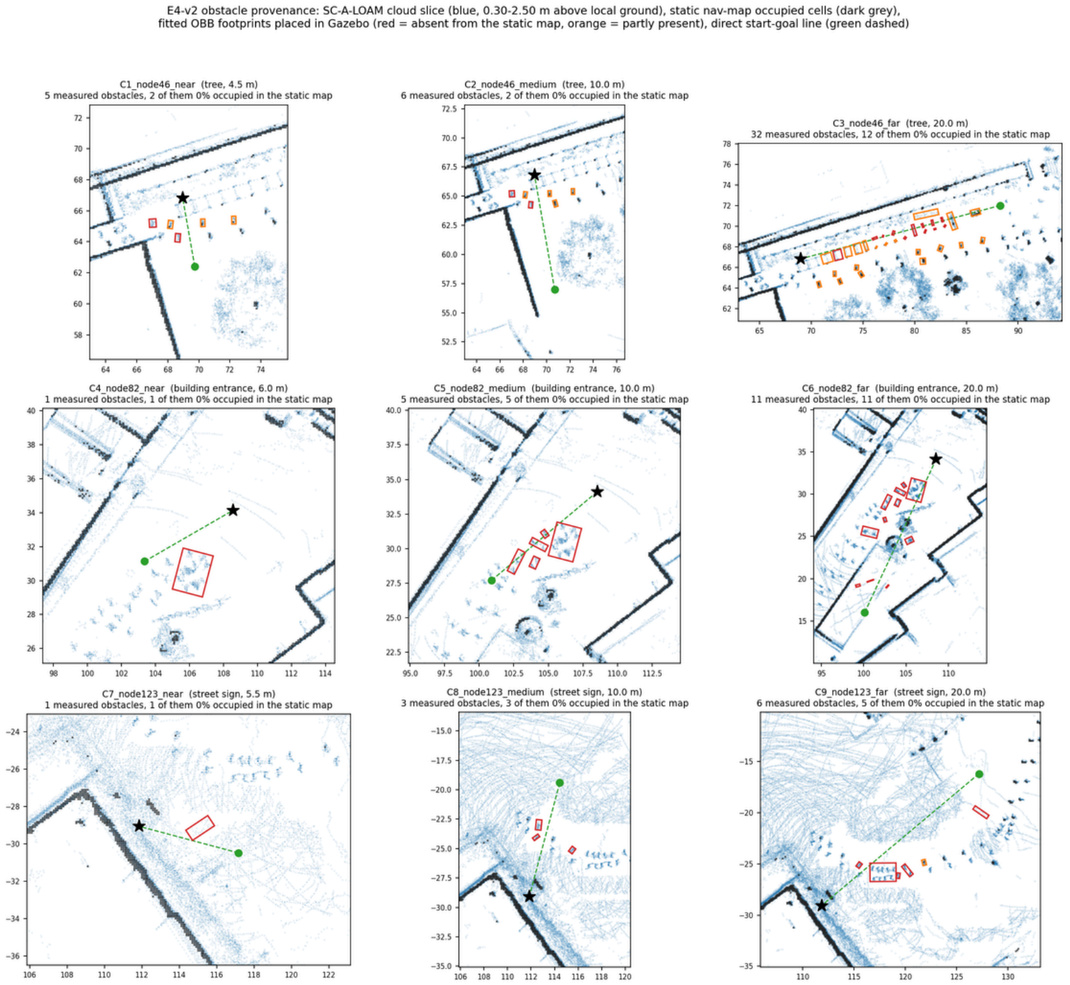
Fallback rule

ray direction d = normalize($[(u_0 - c_x)/f_x, (v_0 - c_y)/f_y, 1]$)

score = distance-to-ray + 0.005 × depth

Select the lowest score; retain only if final ray distance ≤ 0.75 m.

E4 outcomes by target and distance band



Target / band	Success	Min. clr.	Mean
Tree / near	5/5	0.105 m	32.4 s
Tree / med.	5/5	0.167 m	35.7 s
Tree / far	5/5	0.092 m	76.8 s
Entrance / near	5/5	0.525 m	21.5 s
Entrance / med.	5/5	0.106 m	39.5 s
Entrance / far	5/5	0.320 m	70.8 s
Sign / near	5/5	0.506 m	19.3 s
Sign / med.	5/5	0.544 m	34.3 s
Sign / far	5/5	0.545 m	66.5 s

Minimum clearance is the tightest recorded across five repeats.

Evaluation criteria and evidence boundaries

APPENDIX

E1 Mapping	RGB–LiDAR time offset ≤ 0.05 s; report association quality and retained graph evidence.	Supports spatial grounding; the graph remains sparse and calibration-dependent.
E2a Representation	Correct annotated node appears within the evaluated rank; report R@1/5/10 and MRR.	Controlled design-selection evidence on the earlier campus dataset.
E2b Final retrieval	Acceptable node at rank 1 or within ranks 1–5; report top-1, top-5 and MRR@5.	Small 12-input final-pipeline sanity check, not a general benchmark.
E3 Goal and plan	Free cell, clearance ≥ 0.35 m, stand-off 0.8–3.0 m; reachability requires a non-empty plan.	Offline map and three fixed starts; goal validity does not prove connectivity.
E4 Simulation	SUCCEEDED within timeout, XY error ≤ 0.35 m, no collision; Wilson lower bound ≥ 0.70.	Controlled simulation with ground-truth localisation; no sim-to-real safety claim.
E5 Physical robot	SUCCEEDED and XY error ≤ 1.00 m for motion attempts.	Four supervised observations; feasibility and failure-mode evidence only.